MTJ/CMOS-Based CLB Design for Low-Power and CPA-Resistant Secure Nonvolatile FPGA

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***Abstract*—Modern applications such as the Internet of Things (IoT) devices, AI, and automotive applications widely use field-programmable gate arrays (FPGAs). However, many of these applications have limited power resources. Also, the existing FPGAs are vulnerable to side-channel attacks (SCAs) such as correlation-based power analysis (CPA) attacks. Therefore, designing low-power, CPA-resistant, and secure-by-design FPGA is required. In this article, two low-power and CPA-resistant hybrid CMOS/magnetic tunnel junction (MTJ) logic-in-memory-based configurable logic blocks (CLBs) have been proposed and compared to a state-of-the-art counterpart. The first proposed design is single output, and the second one is multioutput. The simulation results show that compared to the state-of-the-art secure CLB counterpart [secured CLB (sCLB) by Zooker et al. (2020)], the proposed CLB designs have 42% and 33% lower delay, 85% and 18% lower power consumption, and 86% and 63% fewer equivalent transistors. To implement one round of the PRESENT algorithm, the first and second designs have 85% and 77% fewer transistors, 42% and 33% lower delay, and 86% and 50% lower power consumption compared to their silicon-proven secure counterpart. Also, to implement convolution layers of binarized neural network (BNN), compared to this counterpart, the first and second proposed designs have 85% and 90% fewer equivalent transistors, 42% and 33% lower delay, and 86% and 79% lower power consumption. Also, the resiliency of the proposed designs against power analysis attacks has been investigated by exhaustive simulations and performing CPA attacks on PRESENT and Advanced Encryption Standard (AES) SBOX. Also, this resiliency has been investigated for different tunnel magnetoresistance ratios (TMRs) and supply voltages.**

***Index Terms*—Configurable logic block (CLB), low-power secure field-programmable gate array (FPGA), magnetic tunnel junction (MTJ), power analysis attack, spintronic.**

I. Introduction

FIELD-PROGRAMMABLE gate arrays (FPGAs) have been widely used in the Internet of Things (IoT) devices, medical applications, AI, and automotive applications [1], [2], [3], [4] mainly because of the higher computing capabilities and flexibility that FPGAs can provide. In addition, infinite reconfigurability and quick design-to-market make FPGAs a promising option to be utilized in these applications. Although FPGAs benefit from these features, due to their limited resources and security concerns in target applications, designing secure and power-efficient FPGAs for these applications is of great importance [1], [2], [4], [5], [6].

Encryption is one of the most used tools to ensure data security in the electronic industry, which needs a secure platform to be implemented. Although many encryption algorithms are theoretically and mathematically secure through the main channel, hardware implementation of these algorithms creates side channels through which information may leak [7], [8], [9]. One particularly effective technique for extracting information through power fluctuation is correlation-based power analysis (CPA). Thus, CPA-resistant implementation of cryptographic algorithms is of great importance [10], [11]. Due to the widespread use of FPGAs in the aforementioned applications and the importance of data security in these applications, designing an inherently secure FPGA platform properly addresses the demand for security while maintaining application flexibility.

Using CMOS-based lookup tables (LUTs) alongside static random-access memory (SRAM) cells forms the configurable logic block (CLB) of the SRAM-based FPGA. SRAM is a volatile memory where stored data are lost after cutting off the power supply. Magnetic tunnel junction (MTJ)-based memory is a nonvolatile memory that can be a promising alternative for SRAM cells in designing CLBs for FPGAs [12], [13]. In the literature, the MTJ is used in designing the CLBs due to their nonvolatility. The nonvolatility of the MTJs eliminates the need for external memory for storing the configuration and reconfiguring the CLBs after the power supply cutoff. Using MTJs alongside novel architecture like logic-in-memory can significantly reduce power consumption, which is vital in applications with limited hardware and power resources [14]. In addition, the manufacturing process of the MTJ is compatible with the CMOS manufacturing process [15], [16].

In the literature, nonvolatile CLBs for FPGAs have been investigated [12], [13], [17]; however, the security of these designs has not been investigated and their read and write operations are vulnerable to side-channel attacks (SCAs). In the literature, two approaches have been used to design secure logic leveraging concealing and hiding approaches. In the concealing approach, power profile randomization is used, which can be in the time domain or amplitude. In [18], a random multitopology logic has been proposed. In [19] and [20], a random delay insertion approach and blurring gates have been proposed, respectively. In the hiding approach, the designers try to eliminate the correlation between power traces and secret information by flattening the power consumption.

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TABLE I
DRAWBACKS OF EXISTING CLB DESIGNS

Design	Technique	Drawbacks
[13]	PCSA	Unsecure against power attacks
[12]	PCSA	Unsecure against power attacks
[21]	Transmission Gate	Unsecure against power attacks
[22]	WDDL	Asymmetric path for out and out' leading to information leakage, high power consumption
[23]	DPMUX	High power consumption

To achieve this goal, dual-rail precharge-based designs are usually utilized. However, when using these designs, some points need to be considered. First, the complementary circuits creating F and F' need to be as symmetrical as possible. Second, the designs should be immune to early evaluation propagation and glitches. Wave dynamic differential logic (WDDL) is considered the most popular dual rail countermeasure for FPGA [22]. However, since the gates in the main path and the complementary path are not the same, the power profile is not completely uniform. In [24], a dual-rail precharge-based multiplexer (DPMUX) has been proposed, in which the gates in the main path and the complementary path are identical. In [23], using the DPMUX, a secure CLB has been proposed and compared to the WDDL-based MUX. The results show that the DPMUX is more secure than WDDL-based MUX. Despite the use of a precharge state in DPMUX, when the device is in evaluation mode, the input can be changed, which leads to a change in the output and power consumption. This can be leveraged by attackers to perform an early evaluation attack on the device. Table I shows the existing designs and their drawbacks. Since the DPMUX-based CLB has the best security compared to other designs, this design has been used as the counterpart for the proposed designs to be compared.

In this article, two secure scalable hybrid CMOS/MTJ LiM-based CLBs have been proposed and compared to a state-of-the-art counterpart. To investigate the power consumption and security of the proposed designs, one round of the PRESENT algorithm, Advanced Encryption Standard (AES) SBOX, and one kernel of a binarized neural network (BNN) have been implemented, and security evaluations have been carried out. The main contribution of this article can be summarized as follows:

1) proposing two CPA-resistant scalable LiM-based CLBs based on a hybrid CMOS/MTJ design;
2) improving area efficiency by enabling the second proposed CLB to have multiple outputs and selector sets;
3) investigating different aspects of functionality performance and reliability of the proposed designs;
4) investigating the resistance of the proposed CLBs against CPA attacks;
5) investigating the effect of tunnel magnetoresistance ratio (TMR) and supply voltage on the resistance of the proposed designs against CPA attack.

The remainder of this article is organized as follows. First, a brief background has been reviewed in Section II. The proposed designs have been discussed in detail in Section III. The functional simulation of the proposed design is presented in Section IV, BNN and the cryptographic benchmark are presented in Section V, and the security of the proposed designs is investigated in Section VI. Finally, Section VII concludes this article.

II. BACKGROUND ON MTJS

Two ferromagnetic layers, separated by a thin oxide layer (barrier), form an MTJ device [25]. The resistance of the MTJ depends on the magnetic orientation of the ferromagnetic layers. One of these ferromagnetic layers has a fixed magnetic orientation which is called a fixed layer. The magnetic orientation of the other ferromagnetic layer can be changed, and it is called the free layer. Based on the relative orientation of ferromagnetic layers, MTJ can have two states called parallel and antiparallel. In the parallel state, the magnetic orientations of the ferromagnetic layers are the same, while in the antiparallel state, the magnetic orientations of the ferromagnetic layers are opposite of each other. The MTJ device shows a relatively low resistance in the parallel state (R_{P}), while it shows a relatively higher resistance in the antiparallel state (R_{AP}). The TMR of an MTJ device indicates the ratio of resistance between two states of the MTJ and is calculated by the following equation:

$$\mathrm{TMR} = \frac{R_{\mathrm{AP}} - R_{\mathrm{P}}}{R_{\mathrm{P}}}. \tag{1}$$

The retention time of the MTJ devices depends on the size of these devices. To use MTJ devices as storage memory, the retention time needs to be high enough. The retention time of MTJ devices can be calculated using the following equation:

$$\tau = \tau_0 \cdot \exp(\Delta), \quad \tau_0 = 1 \text{ ns and } \Delta = \frac{H_{\mathrm{k}} M_{\mathrm{s}} A_{\mathrm{r}} t}{2 k_{\mathrm{B}} T} \tag{2}$$

where Δ is the thermal stability, K_{B} is Boltzmann's constant, T is the system temperature, H_{k} is the uni-axial anisotropy, M_{s} is the saturation magnetization, A_{r} is the area of the MTJ, and t is the thickness of the free layer [26].

Since, in FPGAs, the values of the memory (configuration) are written once and used for a long time (until reconfiguring the FPGA again), the retention time of the MTJs must be long enough to keep the configuration [27], [28], [29]. In order to have a storage class MTJ, the thermal stability of the MTJs must be greater than 75 [14].

III. PROPOSED DESIGN

CLBs generally do not have different selector numbers within a single FPGA device. Instead, they are designed with a predefined number of inputs and multiplexers to route signals within the blocks. The number of inputs and the structure of these multiplexers are fixed and consistent across all CLBs of the same type within the FPGA. This may lead to inefficient use of CLBs since operations with a different number of inputs may be needed. For example, to implement one round of PRESENT, 64 XOR operations followed by 16 SBOX transforms are needed. Assuming that the number of selectors in the CLBs is 4, 64 CLBs are needed to implement the XOR layer, while only two of the selectors of each CLB are used, the configuration leads to significant resource waste. In addition, FPGAs utilize CLBs with different

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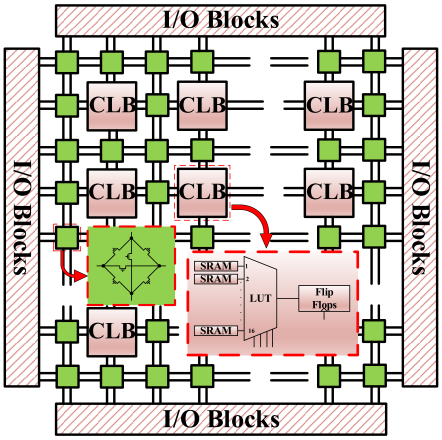

Fig. 1. Structure of conventional FPGA.

numbers of selectors. One of the best ways to build bigger LUTs for CLBs is cascading smaller LUTs. In the case of cascading MTJ-based LUTs, either writing operation on the MTJs (except the MTJs on the first stage) or using output as an input of another CLB is required. Writing and reading operations are vulnerable to differential power analysis (DPA) and CPA attacks. In addition, writing operations with MTJs consume significant power and energy, which is not desirable in applications with limited power constraints [30]. Although different methods of secure implementation of FPGAs have been investigated in the literature, designing a secure FPGA can significantly increase security and reduce the complexity and cost of secure implementations on FPGAs [23]. Also, by reducing the complexity and area overhead of secure implementation on the FPGA, the power consumption will be reduced. Accordingly, in this article, two CLBs have been proposed in which the writing operation has been avoided. As a result, the total power consumption of the circuit has been reduced and the power consumption is more uniform to make the design both low power and resilient against DPA and CPA attacks. The first CLB has been designed in one stage, so the need to write on the MTJs in the further stages has been eliminated. In the second design, MTJ-based LUTs are used in the first stage, and in the further stages, CMOS-based LUTs have been used.

A. First Design

Fig. 1 illustrates a conventional SRAM-based FPGA that consists of I/O blocks, CLBs, and switch boxes. The CLB consists of LUTs, SRAM cells, and flip-flops. Fig. 2 shows the proposed CLB designs and the structure of different components. The first design [see Fig. 2(a)] consists of two components: a 16:1 LUT and a dual rail precharge sense amplifier-based D-flip-flop (PCSA-DFF). The structure of one-stage 16:1 MTJ-based LUT is shown in Fig. 2(c). This LUT consists of a sense amplifier and a multiplexer tree. The configuration will be stored in the MTJs of the multiplexer tree. To store each bit of configuration, two MTJs with opposite states (parallel and antiparallel) are needed. Accordingly, for a 16:1 LUT, 32 MTJs are needed. This LUT has three operation phases: precharge, evaluation, and writing phase. In the precharge phase, the CLK signal is "0" and Write-En is "1"; accordingly, the sense amplifier and multiplexer tree are disconnected by access nMOS transistors, and both sides of the sense amplifier are charged to VDD through precharge pMOS transistors. In the evaluation phase, the CLK and Write-En signals are "1." Two sides of the sense amplifier are connected to a multiplexer tree. Each side faces a resistive path to the ground, and each path has a different resistance value due to the different states of the MTJs. In the write phase, the Write-En signal is "0," and the sense amplifier is disconnected from the multiplexer tree. The write operation will be done using the multiplexer tree. To keep the output of LUT as the output of CLB to be used in the next stages, the proposed dual-rail precharge sense amplifier-based DFF in [31] has been used. This DFF uses signal Sense as the clock signal. When the CLK signal is "1" and after that, the LUT output has been produced, and the DFF enters the precharge mode. This helps the next stage to use the previous stage's output as input. The DFF needs to enter the evaluation phase while the CLK signal is still "1," making sure that the valid outputs are stored in the DFF.

B. Second Design

Fig. 2(b) shows the structure of the second proposed CLB. It consists of four MTJ-based 4:1 LUT in the first layer, four dual-rail precharge sense amplifier-based DFFs in the second layer, a CMOS precharge sense amplifier-based LUT in the third layer, and one dual-rail precharge sense amplifier-based DFF in the last layer. The functionality of the LUTs in this CLB is similar to the first proposed CLB. This CLB has two operational modes: single output and multioutput. In the single-output mode, the first layer LUTs are driven by the same signals as their selector. Hence, the first layer will act as a 16:4 multiplexer. These LUTs use the CLK signal as their clock. The outputs of the first layer are stored in four DFFs as the input of the LUT in the next layer. The first layer of DFFs uses Sense 1 as the clock signal. To complete the multiplexing operation, the CMOS-based LUT in the third stage is used, and its output will be stored in the DFF in the last layer of CLB. The LUT at the third layer and DFF at the last layer use the $\overline{\text{CLK}}$ signal and Sense 2 as their clock. In the multioutput mode, each LUT at the first layer has its own selectors and acts as a 4:1 multiplexer. The outputs of each LUT are stored in the DFFs at the second layer. In this case, the outputs of the DFFs at the second layer are the outputs of the CLB.

Using one-stage LUT will reduce the area overhead and power consumption of one CLB while increasing the delay. On the other hand, using multioutput CLBs will be beneficial if numerous operations with a lower number of operands than the selector of the CLB are needed. In this case, by using the multioutput mode of CLBs, the required function will be implemented on a smaller number of CLBs, which leads to higher power and area efficiency. This feature will be investigated further in Section IV.

Both proposed CLBs in this article are designed to be secure against power analysis attacks and early propagation attacks. To make the designs secure against power analysis attacks, all the components of the CLBs are designed based on a symmetric dual-rail precharge sense amplifier-based logic.

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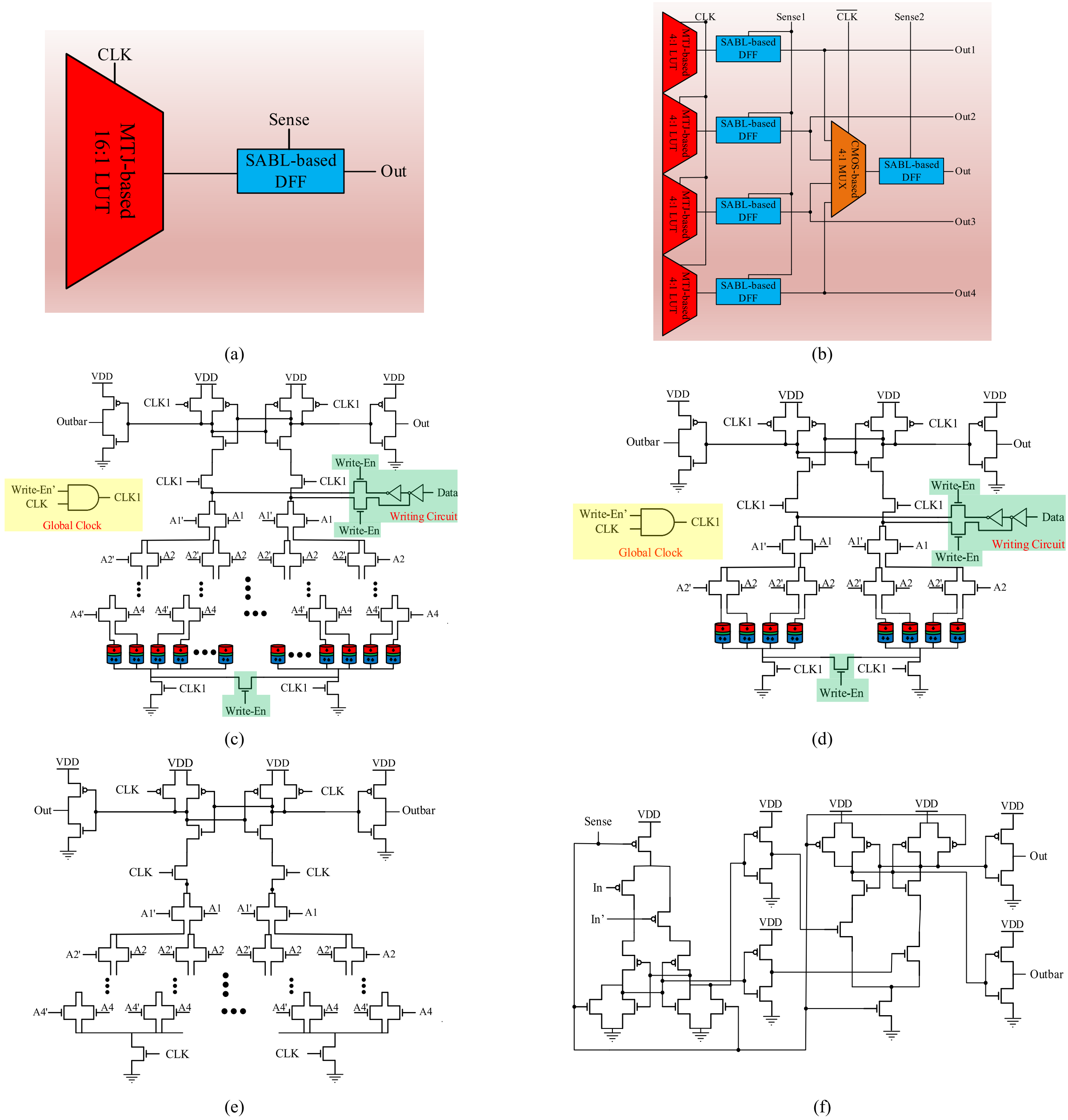

Fig. 2. Proposed designs. (a) First proposed CLB. (b) Second proposed CLB. Schematic of MTJ-based (c) 16:1 LUT and (d) 4:1 LUT. Schematic of (e) precharge-based dual-rail CMOS-based 4:1 MUX and (f) dual-rail precharge-based flip-flop.

This will lead to a uniform power consumption. Uniform power consumption eliminates the correlation between the processed data and the power channel. Accordingly, the attackers cannot find any correlation between these two. In addition, using dual-rail precharge sense amplifier-based logic will eliminate the vulnerability of the proposed CLB to early propagation attacks. Accordingly, the proposed designs provide a secure platform against power attacks and can be easily used in IoT, AI, biomedical, and other applications with limited power and hardware resources.

IV. Simulation

In this section, the simulation and evaluation results of the proposed designs and their counterpart will be investigated and compared comprehensively.

SPICE simulation has been used to simulate the proposed designs and their counterpart. The STT-MTJ model presented in [32] and Taiwan Semiconductor Manufacturing Company (TSMC) 65-nm CMOS process design kit (PDK) have been used for schematic entry and SPICE simulations. It is noteworthy that to have a fair comparison, the counterpart has been optimized and simulated. Table II shows the critical parameters of CMOS transistors and MTJ devices.

Due to the importance of thermal stability and retention time in applications requiring the MTJ to keep its value for a long amount of time, the size of the MTJs is chosen such that they achieve storage class memory standards. For a storage class, memory thermal stability of 75 and higher is needed [14]. The thermal stability and retention time of the MTJs in the proposed CLBs are calculated using (2) and are equal to 77.4 and 4.3 $\times$ 10^{24} s, respectively. This retention time allows

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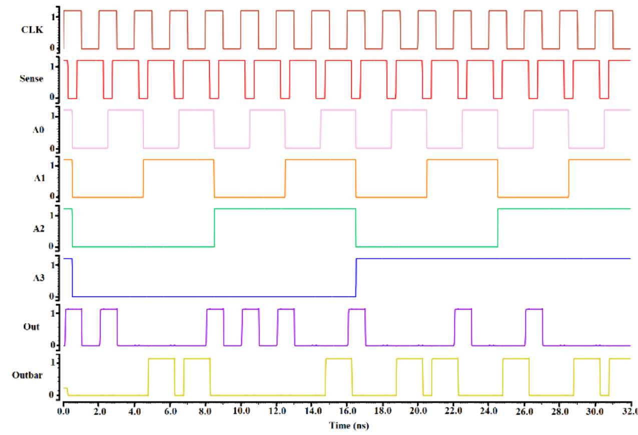
TABLE II
CRITICAL PARAMETERS OF CMOS TRANSISTORS AND MTJS

Description	Value
NMOS	
Gate length	60nm
Gate width	120nm
PMOS	
Gate length	60nm
Gate width	240nm
MTJ	
MTJ surface	60nm×60nm
Oxide barrier thickness	0.85nm
TMR	1.5
Free layer thickness	2nm
Resistance area product	10^{-11}

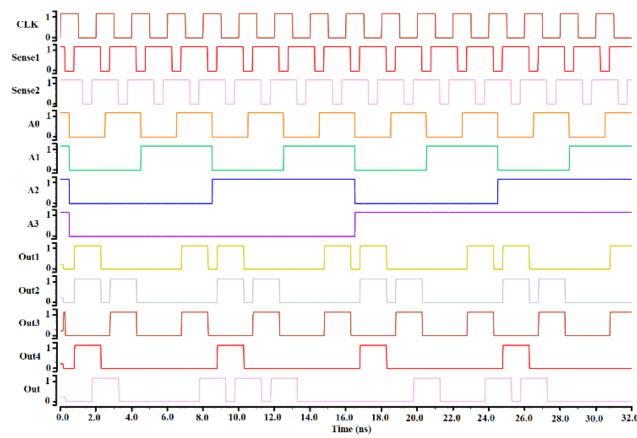
the configuration to be kept for a long time without refreshing it. This will reduce power consumption, which leads to higher battery life.

Fig. 3 shows the output waveform of the proposed CLBs. The output waveform of the first proposed design is shown in Fig. 3(a). The configuration saved on the MTJs in the simulation is “0011000101101011.” When the CLK signal is “0,” the LUT is in precharge mode. When the rising edge of the CLK signal arrives, the LUT will produce the output based on input values and the saved configuration on the MTJs. Since all CLBs use the same clock signal and the LUT will enter precharge mode before the next rising edge of the clock, the output of the LUT needs to be kept for the next stage. Accordingly, the DFF needs to keep its previous output till the LUT produces the output. Hence, the Sense signal needs to go to “0.” After that, the output of the LUT is produced and then goes to “1” before the LUT enters the precharge mode. Then, the Sense signal needs to be steady, so the next stage can use the output of the CLB in the next clock as input. The output waveform of the second proposed CLB is shown in Fig. 3(b). The configuration saved on the MTJ in this simulation is “1001110001011000.” The MTJ-based LUTs and CMOS-based LUTs are in precharge mode when the CLK signal is “0” and “1,” respectively. The MTJ-based LUTs and CMOS-based LUTs enter evaluation mode with rising and falling edges of the CLK signal, respectively. The Sense 1 signal and Sense 2 signal are used as the clock signal of the first layer and output D-FFs. Sense 1 and Sense 2 signals must meet the same criteria as the Sense signal used in the first proposed CLB. The output of the first layer of D-FFs and the output of the DFF at the last stage can be used as input for other CLBs. This can help to reduce the area overhead and the number of used CLBs in situations where many operations with a smaller number of operands than the number of the selectors of the CLBs need to be implemented.

The referring names for the proposed designs and their counterparts are shown in Table III. The simulation results of the proposed CLBs and their counterpart are shown in Table IV. The equivalent unit size transistor (EUST) is used as a metric for comparing area overhead, which is defined as the summation of the number of transistors, each weighted by the ratio of their width to the minimum transistor width [34]. The results show that in case of delay, the first and



(a)



(b)

Fig. 3. Output waveform of (a) first and (b) second proposed designs.

TABLE III
REFERRING NAMES OF THE PROPOSED DESIGNS AND COUNTERPARTS

Referring name	Proposed design in
P1	First design of this paper
P2	Second design of this paper
C1	[23]
C2	[33]
C3	[12]
C4	Conventional CMOS

second proposed designs have 114 and 131 ps and only C4 has a smaller delay compared to the proposed designs. The power consumption of the first and second proposed designs are 7.64 and 42.78 μW. The power consumption of the first proposed design is at least 33% smaller than its counterparts (up to 85% compared to C1). In addition, the power consumption of the second proposed design is 18% smaller than C1 (silicon-proven secure counterpart). Although the power consumption of the second proposed design is higher than C2 and C3, due to the multioutput feature of this design, its power consumption needs more investigation in different benchmarks, which will be investigated in Section V of this article. Also, in the case of the EUST, the first proposed design has an EUST of 106, which is at least 33% lower than its counterparts. Also, while P1, P2, C2, and C3 only use nMOS transistors to implement the LUT, C1 and C4 use both nMOS and pMOS transistors (for pull-up network and transmission

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TABLE IV
PERFORMANCE COMPARISON OF CLBS

Design	P1		P2			C1 [23]		C2 [33]		C3 [12]		C4 (CMOS design)	
	LUT	D-FF	MTJ-based LUT	CMOS-based LUT	D-FF	LUT	D-FF	LUT	D-FF	LUT	D-FF	LUT	D-FF
Delay (ps)	91	23	52	33	23×2	149	24×2	120	24	113	24	88	24
Max frequency (GHz)	8.77		7.63			5.07		6.94		7.29		8.92	
Clock latency	1		1			2		1		1		1	
Power (μW)	7.64		42.78			52.21		11.57		12.59		41.08	
EUST	106		272			742		176		160		402	

gates, respectively), which increases the required number of transistors. In addition, C1 and C4 use SRAM as the memory, while other counterparts use MTJ in a LiM architecture, which increases the number of required transistors for C1 and C4. Accordingly, C1 and C4 have the highest EUST among all the counterparts. Furthermore, due to the structure of the C1, it needs two clocks to produce the output, while the proposed designs in this article produce the output in every clock cycle, which makes the throughputs (clock latencies) of the proposed designs twice more (50% less) than C1 (as the silicon-proven secure counterpart). Cryptographic and BNN benchmarks are used to further investigate the proposed designs' performance and compare them comprehensively with their counterparts.

V. EVALUATION OF CRYPTOGRAPHIC AND BNN BENCHMARK

To better compare and take into account the multioutput feature of the second proposed design, one round of PRESENT cryptography algorithm has been simulated using the proposed design and its counterpart. Also, the estimated result of using these CLBs for implementing the convolution layers of the BNN as a hardware accelerator is given in Table V. This BNN has been used in [28]. This BNN has two convolution layers with 64 masks, two convolution layers with 128 masks, and two convolution layers with 256 masks. All the masks in this BNN are 3×3. By assuming that the input data will enter the circuit in series, and for each feature, only one mask is used, then 8064 XOR operation is needed. The results of these benchmarks are shown in Table V.

A. Cryptographic Benchmark

One round of PRESENT algorithms has been implemented using the proposed designs and their counterpart. As shown in Table V, the power consumption of the first and second proposed designs are 1.075 and 3.764 mW, which compared to the 7.685-mW power consumption of C1 (as the silicon-proven secure counterpart) show 86% and 51% improvement. In addition, the power consumption of the first proposed design is at least 32% lower than its counterparts. Also, the delay of the first and second designs, implementing the PRESENT algorithm, are 0.235 and 0.273 ns, in which only C4 has a lower delay compared to the proposed designs. In addition, for implementing one round of the PRESENT algorithm using the first and second proposed designs 128, 80 CLBs are needed and this number for all the counterparts is 128. The smaller CLB number of the second proposed design is due to its multioutput feature. Furthermore, the first and second proposed designs have EUST of 13 568 and 21 760, which are 85% and 77% fewer than C1 (as the silicon-proven secure counterpart). It is noteworthy that for this benchmark circuit, C2 has a higher EUST compared to the second proposed design, while for a single CLB, it has a lower EUST compared to the second proposed design.

B. BNN Benchmark

The results of implementing a BNN using the proposed designs and their counterpart are also shown in Table V. According to Table V, the power consumption of the BNN for the first and second proposed designs are 61.608 and 86.244 mW, which compared to the power consumption of 421.02 mW of C1 (as the silicon-proven secure counterpart) are 85% and 79% smaller. While the power consumption of the second proposed design CLB is higher than C2 and C3, in the BNN benchmark, its power consumption is lower than all the counterparts. This fact shows that the multioutput feature of the proposed design in some applications can lead to significant power saving. In addition, the hardware overhead of implementing this benchmark using the second proposed design is significantly lower than its counterparts (at least 57.5%). Also, the first proposed design keeps its superiority compared to its counterparts in the case of power consumption and area overhead. Accordingly, both of the proposed designs are promising candidates for implementation of FPGAs for different applications.

According to the results in Table V, choosing the right FPGA with the right CLB is of great importance. This selection can affect the performance of the implemented functions on the FPGA. While selecting the right FPGA, the designer should consider the functions that need to be implemented, the possibility of combining these functions, the average number of selectors of the CLBs, the needed hardware resources, and the available hardware resources on each FPGA. A good selection can lead to better implementation and performance. It is noteworthy that both of the proposed designs can be utilized on the same FPGA chip to increase the performance and efficiency of the FPGA.

By comparing the results of Tables IV and V, it is clear that the first design demonstrates slightly higher efficiency in terms of power consumption and form factor because it is based on the 16:1 LUT. This is useful in applications where more

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TABLE V
SIMULATION RESULTS OF IMPLEMENTING ONE ROUND OF PRESENT ALGORITHM AND BNN

Design		Power (mW)	Delay (ns)	Number of required CLBs	EUST
P1	PRESENT	1.075	0.235	128	13568
	BNN	61.608	306	8064	854784
P2	PRESENT	3.764	0.273	80	21760
	BNN	86.244	352	2016	548352
C1 [23]	PRESENT	7.685	0.413	128	94976
	BNN	421.02	529	8064	5938488
C2 [33]	PRESENT	1.589	0.301	128	22528
	BNN	93.300	386	8064	1419264
C3 [12]	PRESENT	1.783	0.290	128	20480
	BNN	101.525	368	8064	1290240
C4 [CMOS]	PRESENT	5.998	0.237	128	51456
	BNN	331.269	300	8064	3241728

TABLE VI
SIMULATION RESULTS OF POWER CONSUMPTION OF IMPLEMENTING PRESENT SBOX, AES SBOX, AND A BNN KERNEL

DESIGN		P1	P2	C1 [23]	C2 [33]	C3 [12]	C4
NED	PRESENT SBOX	0.1036	0.0617	0.134	0.269	0.305	0.519
	AES SBOX	0.0949	0.0649	-	-	-	-
	3×3 BNN kernel	0.051	0.0284	0.069	0.198	0.243	0.476
NSD	PRESENT SBOX	0.0176	0.0121	0.023	0.126	0.169	0.347
	AES SBOX	0.0206	0.0141	-	-	-	-
	3×3 BNN kernel	0.0012	0.000357	0.0023	0.078	0.108	0.208
Maximum (µW)	PRESENT SBOX	71	267	334	115	131	497
	AES SBOX	1073.2	7514.6	-	-	-	-
	3×3 BNN kernel	78	108	361	131	136	543
Minimum (µW)	PRESENT SBOX	64	250	289	84	91	239
	AES SBOX	971.2	7026.6	-	-	-	-
	3×3 BNN kernel	74	105	339	105	106	284
Average (µW)	PRESENT SBOX	69	259	314	101	113	368
	AES SBOX	1025.6	7237.5	-	-	-	-
	3×3 BNN kernel	77	107	353	116	126	413
STD	PRESENT SBOX	1.214e-6	3.145e-6	7.29e-6	1.28e-5	1.91e-5	1.28 e-4
	AES SBOX	2.119e-5	1.040e-4	-	-	-	-
	3×3 BNN kernel	9.51e-7	3.81e-7	8.21e-6	0.914 e-5	1.37e-5	8.61 e-5

complex functions are needed to map on a single 16:1 LUT. The second design offers enhanced flexibility and optimization for managing hardware resources in applications where less complex functions can be realized on a smaller size 4:1 LUT. Since the second design uses four 4:1 LUTs to design a larger 16:1 LUT, it has four outputs that can be configured as four separate 4:1 LUTs to realize multiple functions. However, if needed, it can also implement a 16:1 LUT. This is advantageous in applications where you need to map multiple functions that can be efficiently realized with the help of 4:1 LUTs.

VI. SECURITY

In this section, the security and resilience of the proposed designs against power analysis attacks have been investigated. PRESENT and AES algorithms are used as benchmarks for SCAs in this article.

PRESENT is a lightweight block cipher with a substitution-permutation network (SPN) structure. It consists of several rounds of substitution (using a small 4 bit to 4-bit SBOX) and permutation (using bitwise rotation). The cryptographic operation is performed on a 64-bit block and the algorithm supports 80 and 128 bits of key. Due to the simplicity of this algorithm and low hardware resource requirements, the PRESENT algorithm is an attractive choice for cryptography in applications with limited resources and power [35].

AES is a block cipher with an SPN structure. AES operates on fixed 128-bit blocks and supports key sizes of 128, 192, or 256 bits. The number of rounds depends on the key size (10, 12, and 14 rounds for 128-, 192-, and 256-bit keys). Each round consists of several steps: SubBytes (a nonlinear substitution using an 8 bit to 8-bit SBOX), ShiftRows (a cyclic shift of rows), MixColumns (a linear transformation), and AddRoundKey (XORing with a round key derived from the main key).

A. Security Metrics

To ensure the security of the proposed design against power analysis attacks, the CLBs have been designed such that they achieve uniform power consumption. Normalized energy deviation (NED) and normalized standard deviation (NSD) have been used as a figure of merit for security against power analysis attacks. Small values of NED and NSD indicate that the power consumption for different inputs is relatively more uniform and the design will be more resilient against

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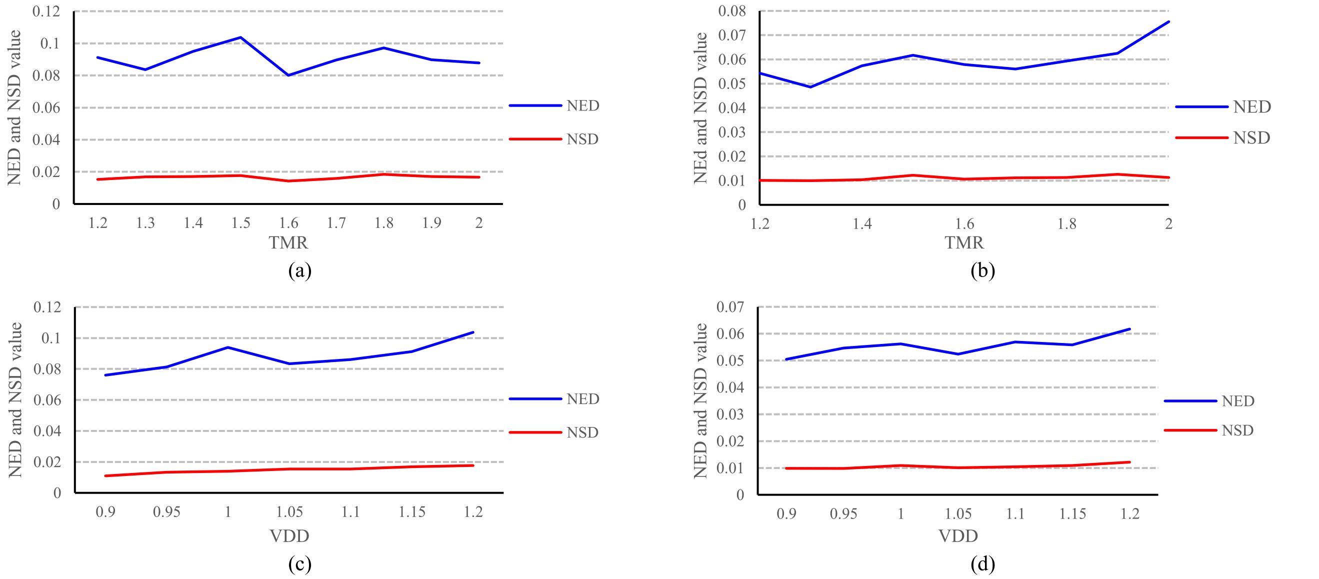

Fig. 4. NED and NSD values of implementation of the PRESENT SBOX with (a) first proposed design with different TMRs, (b) second proposed design with different TMRs, (c) first proposed design with different supply voltages, and (d) second proposed design with different supply voltages.

power analysis attacks. NED and NSD are calculated using the following equations:

$$\mathrm{NED} = \frac{E_{\max} - E_{\min}}{E_{\max}} \tag{3}$$

$$\mathrm{NSD} = \frac{\sigma_{\mathrm{E}}}{\mu_{\mathrm{E}}}. \tag{4}$$

1) Security Metrics for Cryptography and BNN Applications: Table VI shows the NED and NSD of implementing PRESENT SBOX, AES SBOX, and a 3×3 BNN kernel. Also, the minimum, maximum, average, and standard deviation of the 5000 gathered power traces are shown in Table VI. The NED and NSD values of the first proposed design for PRESENT SBOX are 0.1036 and 0.0176, respectively, and these values for the second proposed design are 0.0617 and 0.0121, respectively, which are smaller than these values for C1 (0.134 and 0.023, respectively) as the silicon-proven secure counterpart. Also, NED and NSD values of implementing the BNN kernel are 0.051 and 0.0012, respectively, and these values for the second proposed design are 0.0284 and 0.0141, respectively, while these values for C1 are 0.069 and 0.0023, respectively. As shown in Table VI, both of the proposed designs have better NED and NSD compared to their counterparts, and also, these values are relatively small, which indicates that their power consumption for different inputs is relatively uniform and the proposed designs have good resilience against power analysis attacks. For further security investigation of the proposed designs against power analysis attacks, CPA attacks on PRESENT SBOX have been performed.

a) Security metrics comparison of AES and PRESENT SBOX: In addition to PRESENT SBOX, which is a 4-bit SBOX, an 8-bit AES SBOX has been implemented using the proposed design. This study aims to investigate the security of the proposed designs for implementing different functions with different sizes.

The NED and NSD values of the first proposed design implementing PRESENT SBOX are 0.1036 and 0.0176, respectively, while these values for implementing AES SBOX using the first proposed design are 0.0949 and 0.0206, respectively. In addition, the NED and NSD values for implementing the PRESENT SBOX using the second proposed design are 0.0617 and 0.0121, respectively, and these values for implementing AES SBOX using the second proposed design are 0.0649 and 0.0141, respectively. The difference between NED and NSD values for implementing PRESENT and AES SBOX are 0.0046 and 0.003, respectively, and these values for the second proposed design are 0.0032 and 0.002, respectively, which are negligible and indicate that the power consumption of the proposed designs, independent of the implemented function and its size, is relatively uniform and the proposed designs keep their resiliency against power analysis attack.

2) Security Metrics for Different TMRs and Supply Voltages: To study the effect of TMR and supply voltage on the security of the proposed designs, a PRESENT SBOX has been simulated using the proposed designs with different TMR and supply voltages. The results are shown in Fig. 4. As the results show, the NED and NSD values for different TMR and different supply voltage values are very small, the NSD values are almost equal, and the NED values have very small fluctuations. The NED values for the first proposed design are in the range of 0.08 and 0.103 and in the range of 0.076 and 0.103 for different TMR and VDD, respectively. For the second proposed design, the values of NED are in the range of 0.048 and 0.075 and in the range of 0.05 and 0.061 for different values of TMR and VDD, respectively. Also, the NSD values for the first proposed design are in the range of 0.014 and 0.018 and in the range of 0.01 and 0.017 for different values of TMR and VDD, respectively. The NSD values for the second proposed design are in the range of 0.0099 and 0.012 and in the range of 0.0098 and 0.012 for different values of TMR and VDD, respectively. Due to the negligible difference between the NSD and NED values and their small values, it can be concluded that the security of the proposed design against power attack is almost independent

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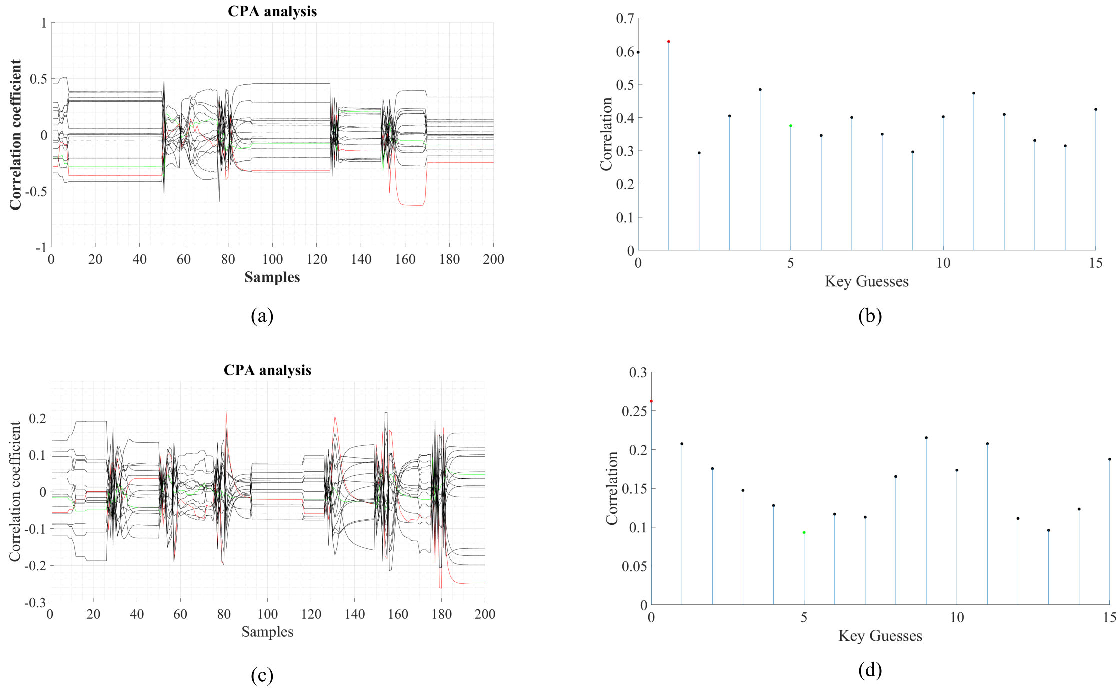

Fig. 5. Correlation coefficients of the power consumption and different keys for PRESENT SBOX, implemented using (a) and (b) first proposed design and (c) and (d) second proposed design.

of the values of TMR and supply voltage and the proposed designs are relatively resilient against power analysis attack.

B. Power Analysis Attack

DPA attack is a sophisticated and powerful attack that takes advantage of variations in power consumption as the device performs cryptographic algorithms or other sensitive tasks [36]. In a DPA attack, the attacker captures power traces of the device and tries to find the correlation between the device's power consumption and its internal operation by exploiting the fact that different input data values and operations can result in different power consumption patterns. Like a DPA attack, CPA is a noninvasive attack method that leverages correlations between the power consumption of a device and its internal operations by trying to find the correlation between the power traces and reference traces [37].

Both DPA and CPA are proven to be powerful, but CPA needs fewer power traces to be successful if the device is attackable [38]. Accordingly, a CPA attack has been performed to ensure the security of the proposed designs against power analysis attacks. It is worth mentioning that the proposed design in [23] has been proven to be secure against power analysis attacks. Meanwhile, the simulation results show that the proposed designs have more security against power analysis attacks compared to the proposed design in [23], according to their lower values for NED and NSD.

The attack hypothesis model used in this article is as follows: the Hamming distance model has been used as the information leakage model for CPA attacks. For both the PRESENT and AES algorithms, the registered output of one SBOX at the first round of encryption and the preceding XOR operation has been chosen as the point of interest (POI). In addition, all environmental noise was eliminated, and only one SBOX and the preceding XOR operations have been attacked. It is noteworthy that by attacking only one SBOX (and not one round), the algorithmic noise caused by concurrent cryptographic operations was minimized, enabling the authors to evaluate the security of the proposed design under a worst-case scenario. This approach assumes that the attacker has access to noise-free power traces, full knowledge of the cryptographic algorithm, and plaintext data, with the secret key being the only unknown.

1) CPA Attack on PRESENT SBOX: To perform the CPA attack, a PRESENT S-box and four XOR operations have been simulated using Cadence Specter. These simulations have been carried out at the frequency of 500 MHz in a noise-isolated environment. It is noteworthy that 5000 power traces have been gathered. Also, the frequency of the sampling is 100 GHz (200 samples per clock), the VDD is 1.2 V, and other parameters of the simulation are set based on the reported numbers in Table II. Fig. 5 shows the unsuccessful attack with the key value of 5 for the first and second proposed designs. The correlation coefficients of all possible keys are shown in Fig. 5(a) and (c), and Fig. 5(b) and (d) shows the maximum correlation of each possible key. The red line in Fig. 5(a) and (c) shows the correlation coefficient of the guessed key, and the green line is the correlation coefficient of the actual key. The actual key for both of the proposed designs has been set to 5, and the guessed key of the CPA attack for the first and second designs is 1 and 0, respectively. Fig. 5 shows that the CPA attacks with 5000 power traces on both first and second designs were unsuccessful, which was predictable based on the calculated values of NED and NSD for the proposed designs.

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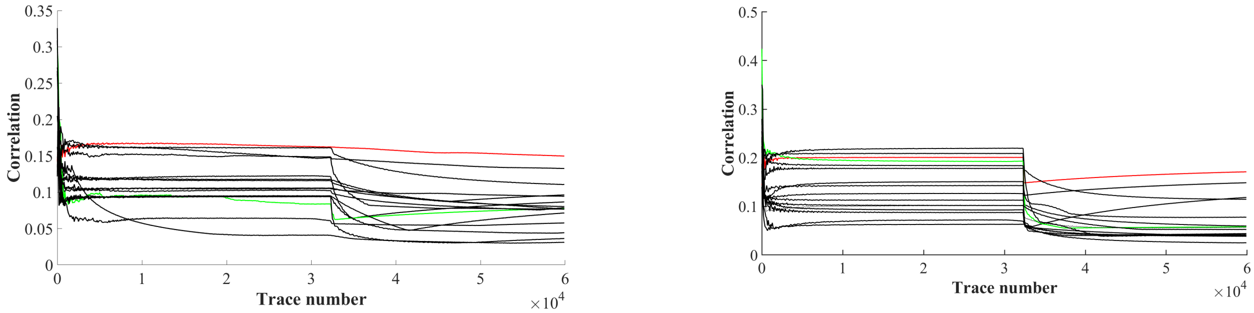
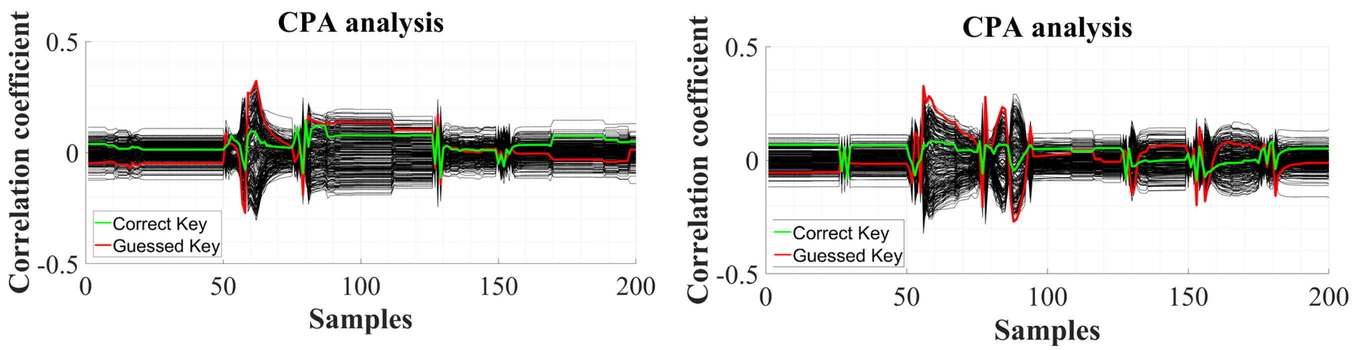


(a) (b)

Fig. 6. Correlation of the power consumption and each key for different numbers of power traces. SBOX utilizing (a) first proposed design and (b) second proposed design.



(a) (b)

Fig. 7. Correlation coefficients of the power consumption of the AES SBOX, implemented using (a) first proposed design and (b) second proposed design.

For further investigation of the resiliency of the proposed designs against the CPA attack, 60 000 power traces for each design have been gathered. Fig. 6 shows the results of the CPA attack with 60 000 power traces on both of the proposed designs. The results show that both of the proposed designs are resilient against CPA attacks.

2) CPA Attack on AES SBOX: In order to investigate the resiliency of the proposed designs against CPA attack for larger circuits, the CPA attack was performed on AES SBOX that was implemented using the proposed designs. In order to gather the power traces for this attack, a frequency of 500 MHz, a sampling frequency of 100 GHz, and a VDD of 1.2 V have been utilized for simulations. Also, other critical parameters have been set as the values in Table II.

The unsuccessful CPA attack on the first and second proposed designs, implementing the AES SBOX, is shown in Fig. 7. The red line in Fig. 7 shows the correlation coefficient of the guessed key, and the green line is the correlation coefficient of the actual key. The actual key for both of the proposed designs has been set to 0, and the guessed key by CPA attack for the first and second designs is 192 and 144, respectively. The unsuccessful attacks confirm the resiliency of the proposed designs against the power analysis attack, which has been indicated by the small values of NED and NSD. In addition, the unsuccessful attacks on the AES SBOX show that the proposed designs, independent of the functions they are implementing, are resilient against power analysis attacks. Accordingly, utilizing the proposed designs can provide an algorithm-independent power analysis attack-resilient platform for implementing different functions.

The values of NED and NSD and the results of performing CPA attacks show that both proposed designs have good resilience against power analysis attacks. The resilience of the proposed designs against power analysis (as the most common SCA) alongside the low power consumption of these designs makes them a promising candidate for designing low-power secure FPGAs, which is of great importance in applications such as IoT devices, AI, and automotive applications.

VII. Conclusion

In this article, two low-power nonvolatile spintronic-based CLBs based on MTJ have been proposed. The proposed designs are proven to be secure against CPA attacks. Furthermore, in comparison with the state-of-the-art counterpart, the proposed designs have lower delay, power consumption, and less transistor number. Also, the implementation of one round of PRESENT cryptography algorithm, AES SBOX, and convolution layers of a BNN has shown the low power consumption and CPA resiliency of the proposed MTJ/CMOS-based CLBs. The results of NED and NSD values for implementing the PRESENT SBOX and AES SBOX show that the proposed designs, independent of the implemented functions on them, are resilient against power analysis attacks. Also, it is shown that the resiliency of the proposed designs against CPA attacks will be consistent in the case of using

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MTJs with different TMR ratios and different supply voltage values. It is worth mentioning that the designers can utilize different CLBs (single-output or multioutput) based on the functions and algorithms they intend to implement. The designers can use both proposed designs in this article by considering the features of each design and the algorithms and functions they intend to implement. The incorporation of a multioutput feature in the second proposed design allows the implementation of more than one function on the same CLB, which leads to higher resource allocation efficiency. In addition, utilizing FPGAs with both of the proposed designs would not only enhance hardware efficiency but also provide greater flexibility in adapting to varying application demands. This is possible by leveraging the lower power consumption of implementing more complex functions with the first proposed design and the lower hardware overhead of implementing functions with lower complexity using the second proposed design. We believe that the proposed design of low-power and CPA-resistant MTJ/CMOS-based CLBs can serve as a starting point for designing low power and secure by design nonvolatile FPGAs.

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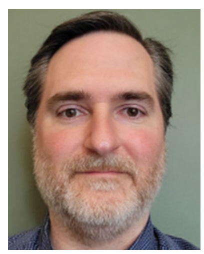

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